

LETTER

Depth profiles of Jerlov water types

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Scientific Significance Statement

The clarity of the world's oceans ranges from very clear open ocean water through to turbid coastal waters. The Jerlov ocean classification is a useful tool that groups ocean waters into 10 types based on their near-surface clarity. However, the clarity of ocean water varies with depth. Technologies such as optical communications, and processes such as photosynthesis, rely on the vertical transmission of light through ocean waters, and therefore are affected by this depth-dependence. This study has used a vast database of ocean optical measurements to derive a new ocean classification across 20 depth layers between 0 and 200 m. This classification will find use in applications where modeling of typical ocean waters requires the depth-dependence of clarity to be accounted for.

Abstract

Typical depth profiles of Jerlov water types have been derived to characterize the clarity of the world's oceans. Measured values of the downwelling diffuse attenuation coefficient, taken from a world-wide database, were quantitatively analyzed. Depth profiles were extracted from more than 2500 data collection campaigns, consisting of the closest-matching Jerlov water types at the uppermost layer (0–10 m) and the corresponding Jerlov water types at deeper depth layers down to 200 m. A table of “typical” depth profiles for the 10 Jerlov water types was generated based on the maximum campaign count at each depth layer. This new ocean classification will find use in applications where modeling of typical ocean waters requires the depth-dependence of clarity to be accounted for.

Jerlov water types are a convenient method of classifying the world's oceans by their near-surface clarity. First defined in 1951 (Jerlov 1951), there are 10 Jerlov water types to characterize oceans from the clearest Jerlov I to the most turbid Jerlov 9C (I, IA, IB, II, III, 1C, 3C, 5C, 7C, 9C). These water types are quantified by their downwelling diffuse

attenuation coefficient, K_d (m^{-1}), from 310 to 700 nm (Jerlov 1976) (see Fig. 1). K_d is an apparent optical property that is calculated as the rate of change of the natural logarithm of downwelling irradiance with depth. Whilst K_d is most sensitive to absorption, it also includes loss due to scattering. Jerlov's classification is based upon measurements of

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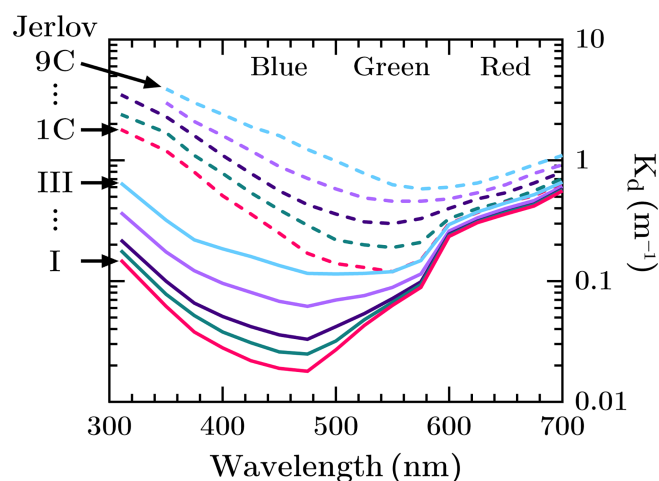


Fig. 1. Downwelling diffuse attenuation coefficient, K_d (m^{-1}), for each of the 10 Jerlov water types (Jerlov 1976).

K_d in the first 10 m of water depth due to the homogeneous composition in this region.

Some applications require an understanding of ocean clarity at depths beyond those defined by the Jerlov water types. Optical communications (Chen et al. 2022) and underwater imaging (Vlachos and Skarlatos 2021) can both involve light propagation at depths below the 10 m at which the Jerlov classification is defined. Furthermore, light penetration through deeper ocean depths affects biological processes such as photosynthesis (Kirk 1994; Falkowski and Raven 2007). As ocean clarity can vary with depth (Smart 1993), this needs to be accounted for when considering such cases.

Several studies have measured the vertical profile of clarity in specific oceans. Jerlov and Koczy (1951) found that a particularly clear part of the North Atlantic exhibited higher turbidity from 100 to 200 m compared with the upper 100 m, while a region of the South Pacific was found to have a relatively uniform vertical profile of clarity. Smart (1993) studied a similar area of the North Atlantic and found that the chlorophyll concentration increased with depth down to around 100 m, which resulted in an associated increase in the diffuse attenuation coefficient. More turbid oceans were found to have higher chlorophyll concentrations (and hence higher K_d) in the upper layers, which substantially decreased below around 50 m. Several authors have explained the increase in K_d with depth for clear waters by the presence of a so-called “deep chlorophyll maximum” (Bouman et al. 2000; Piazena et al. 2002; Souto et al. 2007; Xiu et al. 2009).

At present, there are no “typical” depth profiles available to characterize the clarity of the world’s oceans. The works referenced above were confined to a small number of regions where measurements were taken in situ. Jerlov (1976) did present some limited data on depth profiles for the Jerlov water types at 465 nm (see Discussion below). Although Jerlov recognized that extending his ocean classification to greater depths would be

useful, he cited vertical stratification as one reason this may be challenging. There have been no other published attempts to define the vertical profile of typical ocean waters in this way.

The present study aims to derive “typical” depth profiles for the Jerlov water types. We have used the World-wide Ocean Optics Database (WOOD) (Smart 2000, 2012) as a geographically diverse source of measured data for K_d at different depths. We quantitatively analyzed this vast dataset and extracted depth profiles that begin with the Jerlov water types at the uppermost layer, and then we used the same Jerlov water types to describe the clarity at different depth layers. The resulting profiles will find application where modeling of typical ocean waters requires the depth-dependence of clarity to be accounted for.

Methods

The data analysis method is similar to the one we developed previously to derive inherent optical properties for Jerlov water types (Williamson and Hollins 2022), and we have provided the complete dataset with source code (Williamson and Hollins 2023). Our data source is WOOD, which provides a single data file with experimentally measured values for K_d from the world’s oceans across hundreds of cruises and research teams, and spanning several decades. Each line of the data file (known as a “cast”) holds an array of depths with corresponding measured values of K_d at one specified wavelength, and the raw file has 141,039 casts containing a total of 5,347,115 measurements. We have processed these data using the Python programming language version 3.9.4 with the pandas data analysis library version 1.2.4 (Reback et al. 2021).

The WOOD data were cleaned to provide a robust dataset for the subsequent analysis. All K_d values indirectly derived from chlorophyll measurements were removed, as such values are predominantly valid only for open ocean waters and shallow depths (Morel et al. 2007). This left only K_d values measured directly from sunlight penetration at specific wavelengths and specific depths. All measurements flagged as having questionable quality or provenance were also removed. Next, we removed all measurements at wavelengths greater than 500 nm to leave only those measurements at shorter wavelengths where there is the greatest difference between K_d values (see Fig. 1). This filtering would improve the accuracy of Jerlov classification at a later stage. Finally, casts were removed that did not hold measurements at depths of 10 m or less, as those depths were required for initial classification of Jerlov water types. A total of 22,876 casts remained after the cleaning, with a total of 1,613,338 measurements.

For each cast, the average value of K_d across 20 depth layers was calculated. These layers were in 10 m increments from 0 to 200 m (i.e., 0–10, 10–20 m, etc.), with the deeper bound being inclusive for a given layer (e.g., 0–10 m included measurements at 10.0 m, which were excluded from the 10–20 m layer). By design of the data cleaning, all 22,876 casts contained data across 0–10 m, while 92% of these casts held data from 10 to

20 m, reducing to 28% from 100 to 110 m, and 10% from 190 to 200 m. Only 1.5% of casts held data at depths greater than 200 m and these did not provide sufficient data points for robust subsequent analysis, so they were removed.

Casts were then grouped into data collection “campaigns,” defined as a group of casts from the same cruise that were taken at approximately the same time (up to 12 h after the first cast of a campaign) and location (within $\pm 0.5^\circ$ of latitude and longitude from the first cast of a campaign). Where measurements existed for the same wavelength in the same campaign these were averaged. This grouping resulted in 2987 campaigns with an average of 3.2 wavelengths per campaign (standard deviation of 2.2 wavelengths, minimum of 1, maximum of 9, and median of 3, with a total of 34 unique wavelengths from 305 to 495 nm).

Jerlov classifications were applied to each campaign for each depth layer. Classification was achieved for a given campaign and depth layer by comparing the full spectrum of available K_d values to the corresponding K_d values for each

Jerlov water type and finding the best fit (only matches within 25% were kept for the subsequent analysis). The final result was 2571 campaigns with a Jerlov classification across 0–10 m, with Fig. 2 showing the diversity of these locations on a world map. 87% of these campaigns had corresponding Jerlov classifications from 10 to 20 m, reducing to 37% from 100 to 110 m, and 7% from 190 to 200 m.

Results

Table 1 shows the results of the analysis for six of the 20 depth layers: the number of campaigns matched to each Jerlov water type at each depth layer, referenced against each near-surface Jerlov water type (please see our dataset [Williamson and Hollins 2023] for the full analysis across all 20 layers). The table shows that there is no single “standard” depth profile for each of the near-surface Jerlov water types. However, it can be seen that there are strong trends, with one or two Jerlov water types dominating any given depth layer

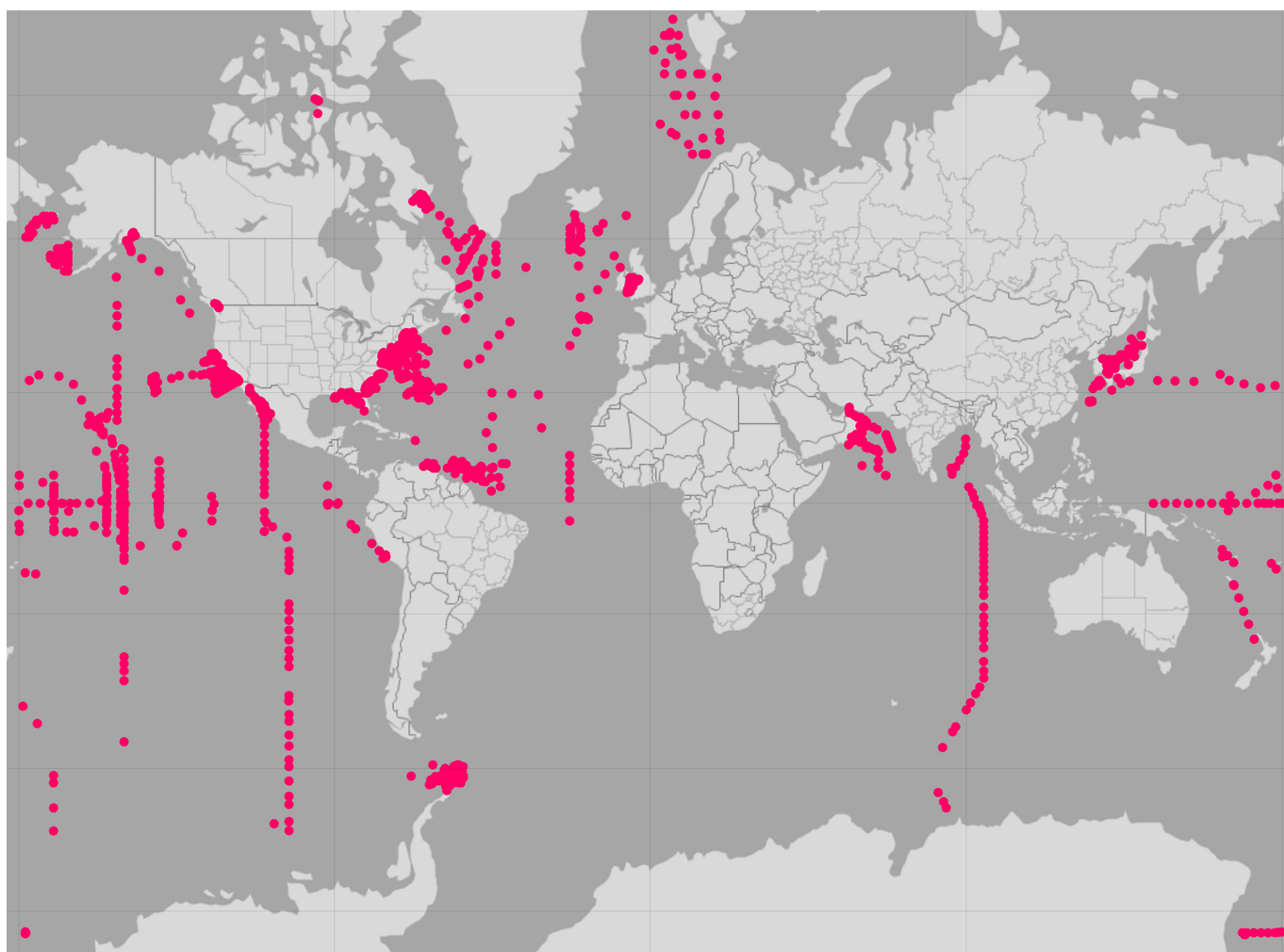


Fig. 2. World map showing the 2571 campaign locations for K_d measurements.

Table 1. Number of campaigns matched to each Jerlov water type at six of the 20 depth layers, referenced against each near-surface Jerlov water type.

0–10 m near-surface layer		I	IA	IB	II	III	1C	3C	5C	7C	9C
Depth layer		Number of matching campaigns									
10–20 m	I	118*	41	14	5	0	0	0	0	0	0
	IA	70	161*	76	16	1	0	1	0	0	0
	IB	15	92	311*	36	2	0	1	0	0	0
	II	6	4	36	416*	20	1	2	0	1	0
	III	0	0	2	25	320*	15	7	1	0	0
	1C	0	0	0	1	39	174*	33	4	1	0
	3C	0	0	0	0	6	21	79*	15	5	0
	5C	0	0	0	0	0	1	8	18*	5	1
	7C	0	0	0	0	0	0	0	2	11*	1
50–60 m	9C	0	0	0	0	0	0	0	0	1	0
	I	5	9	4	2	1	1	2	0	0	0
	IA	53	48	25	7	4	0	0	0	0	0
	IB	96*	167*	199*	47	19	13	5	0	0	0
	II	26	31	110	319*	217*	91*	28*	0	0	0
	III	1	7	11	27	31	20	6	1	1	0
	1C	0	1	0	3	10	6	1	0	0	0
	3C	0	0	1	2	2	1	6	0	0	0
	5C	0	0	0	0	0	0	0	0	0	0
100–110 m	7C	0	0	0	0	0	0	0	0	0	0
	9C	0	0	0	0	0	0	0	0	0	0
	I	1	2	0	2	5	0	0	0	0	0
	IA	7	9	14	10	7	4	0	0	0	0
	IB	64*	113*	208*	100*	30	13*	0	0	0	0
	II	56	82	70	99	36*	5	3	1	0	0
	III	0	1	0	0	2	0	1	0	0	0
	1C	0	0	0	1	0	0	1	0	0	0
	3C	0	0	0	0	0	0	2	0	0	0
150–160 m	5C	0	0	0	0	0	0	0	0	0	0
	7C	0	0	0	0	0	0	0	0	0	0
	9C	0	0	0	0	0	0	0	0	0	0
	I	0	2	2	6	2	0	0	0	0	0
	IA	15	18	38	11	4	2	1	0	0	0
	IB	55*	86*	120*	66*	12*	4	0	0	0	0
	II	1	15	21	35	5	0	0	0	0	0
	III	1	1	0	3	0	0	0	0	0	0
	1C	0	0	0	0	0	0	0	0	0	0
190–200 m	3C	0	1	0	0	0	0	0	0	0	0
	5C	0	0	0	0	0	0	0	0	0	0
	7C	0	0	0	0	0	0	0	0	0	0
	9C	0	0	0	0	0	0	0	0	0	0
	I	0	4	2	2	0					
	IA	14	23*	19	7	1					
	IB	16*	22	37*	11*	1					

(Continues)

Table 1. Continued

0–10 m near-surface layer		I	IA	IB	II	III	1C	3C	5C	7C	9C
Depth layer		Number of matching campaigns									
	II	1	3	5	0	1					
	III	0	0	1	0	2					
	1C	0	1	0	0	0					
	3C	0	0	0	0	0					
	5C	0	0	0	0	0					
	7C	0	0	0	0	0					
	9C	0	0	0	0	0					

*The maximum value for each combination (for maxima ≥ 10).

Table 2. The typical depth profile of Jerlov water types as derived from this study, showing the water type at each depth layer with the maximum campaign count from this study. The top row (0–10 m) shows the near-surface Jerlov water type.

Depth layer		Jerlov water type									
0–10 m	I	IA	IB	II	III	1C	3C	5C	7C	9C	
10–20 m	I	IA	IB	II	III	1C	3C	5C	7C		
20–30 m	IA	IA	IB	II	III	III	III				
30–40 m	IA	IB	IB	II	III	III	III				
40–50 m	IB	IB	IB	II	II	II	II				
50–60 m	IB	IB	IB	II	II	II	II				
60–70 m	IB	IB	IB	II	II	II	II				
70–80 m	IB	IB	IB	II	II	II					
80–90 m	IB	IB	IB	II	II	II					
90–100 m	IB*	IB	IB	II	II	IB					
100–110 m	IB	IB	IB	IB	II	IB					
110–120 m	IB	IB	IB	IB	II						
120–130 m	IB	IB	IB	IB	IB						
130–140 m	IB	IB	IB	IB	IB						
140–150 m	IB	IB	IB	IB	IB						
150–160 m	IB	IB	IB	IB	IB						
160–170 m	IB	IB	IB	IB	IB						
170–180 m	IB	IB	IB	IB							
180–190 m	IB*	IB	IB	IB							
190–200 m	IB	IB*	IB	IB							

*Cases where the second-highest campaign count was chosen as this count was close to the maximum and more consistent with the surrounding depth layers.

for a given near-surface Jerlov water type. For example, for an ocean classified as Jerlov IB from its near-surface (0–10 m) K_d measurements, looking at the 100–110 m depth layer shows that 208 campaigns matched as Jerlov IB and 70 matched as Jerlov II, which were both significantly higher than any other water type.

The purpose of this study was to derive “typical” depth profiles for Jerlov water types, and therefore the water type with the maximum campaign count at each depth layer was declared to be the representative “typical” water type. In three out of the 119 cases in Table 2, a “typical” water type declaration was made on the second-highest water type where the

campaign counts were close and it was more consistent with the surrounding layers. A minimum campaign count of 10 was enforced for the declarations, in order to remove any cases with sparse data.

Table 2 shows the resulting “typical” depth profiles for the 10 near-surface Jerlov water types. Complete (0–200 m) depth profiles have been derived for the four Jerlov water types from I to II, while Jerlov III was declared down to 170 m, Jerlov 1C down to 110 m, Jerlov 3C to 70 m and Jerlov 5C and 7C down to 20 m. Jerlov 9C could not be declared below the uppermost layer due to the lack of available measurements.

Discussion

Typical depth profiles have been derived for Jerlov water types, and they exhibit trends matching those found in the literature. From Table 2 it can be seen that all water types from Jerlov I to Jerlov 7C maintain their near-surface (0–10 m) Jerlov classification down to 20 m depth. Clear waters (Jerlov I and Jerlov IA) then degrade in clarity with depth, until they reach Jerlov IB at 40 m and below for near-surface Jerlov I, and at 30 m and below for near-surface Jerlov IA. Jerlov IB maintains a constant clarity from the near-surface to 200 m. The other water types typically improve in clarity with depth, with near-surface Jerlov II, III and 1C all tending to the profile of Jerlov IB (at 100 m and below for near-surface Jerlov II, 120 m and below for near-surface Jerlov III, and 90 m and below for near-surface Jerlov 1C). Jerlov 3C improves in clarity with depth until it reaches Jerlov II at 40 m and below.

The initial degradation in clarity with depth for Jerlov I and Jerlov IA aligns with the findings of Jerlov and Koczy (1951) and Smart (1993) from measurements in clear areas of the North Atlantic. This degradation is explained by the presence of a “deep chlorophyll maximum” in clear waters, which results in increased attenuation with depth until the chlorophyll peak is passed. Beyond this chlorophyll peak, it is possible that colored dissolved organic matter (CDOM) begins to dominate attenuation and compensate for the reduction in chlorophyll, hence maintaining a consistent clarity (Westberry et al. 2008), although depth profiles of CDOM are known to vary considerably (Nelson and Siegel 2013). This characteristic is not seen in more turbid waters in the present study, which is also supported by the findings of Smart who reported higher chlorophyll concentrations in the upper layers for turbid waters. Uitz et al. (2006) and Westberry et al. (2008) have published depth profiles of chlorophyll and related parameters. Uitz showed that the depth profile of chlorophyll correlates strongly with its concentration at the surface across a wide variety of locations. Westberry describes seasonal variations in the profiles of chlorophyll and K_d , and finds evidence that CDOM absorption becomes stronger in the deeper layers. The profiles reported in both papers exhibit several features consistent with our own study.

The results from this study also provide a good match to the only other published data that could be found on depth

Table 3. The depth profile of Jerlov water types as derived from data published by Jerlov (1976). The top row (0–10 m) shows the near-surface Jerlov water type.

Depth layer	Jerlov water type					
0–10 m	I	IA	IB	II	III	1C
10–20 m	I	IA	IB	II	III	III
20–30 m	IA	IA	IB	II	III	III
30–40 m	IA	IB	IB	II	III	III
40–50 m	IA	IB	IB	II	III	
50–60 m	IB	IB	II	II		
60–70 m	IB	IB	II	II		
70–80 m	IB	IB	II	II		
80–90 m	IB	IB	II			
90–100 m	IB	IB	II			
100–110 m	IB	IB	II			
110–120 m	IB	IB				
120–130 m	IB	IB				
130–140 m	IB	II				
140–150 m	IB					
150–160 m	IB					

profiles for Jerlov water types. Jerlov (1976) published a graph of downward irradiance percentage vs. depth (0–160 m) for eight Jerlov water types (I, IA, IB, II, III, 1C, 5C, and 9C) at 465 nm (fig. 71 in the reference). The source of the data was not specified and its provenance is unknown. We converted these data into K_d and identified the closest Jerlov water type at each of our depth layers to derive the results in Table 3 for Jerlov I to 1C (please see our dataset (Williamson and Hollins 2023) for the full data). While the Jerlov data cover fewer water types and depths than the present study, they provide a useful comparison nonetheless. The Jerlov data show a clarity degradation with depth at Jerlov I and IA, with almost identical profiles to those derived by this study. However, there is also a clarity degradation with depth at Jerlov IB, while Jerlov II and Jerlov III remain constant, and Jerlov 1C improves with depth.

It is important to note that we selected the maximum campaign counts to represent “typical” depth profiles, but this is not an accurate depth profile for all such near-surface water types, and it does not account for any seasonable variability. In the full results that accompany this paper we have provided additional parameters alongside each campaign count to permit a finer analysis of their veracity. The first of these parameters is the total number of different cruises represented by the matched campaigns, with a higher cruise count indicating a more diverse range of results. Second, the number of unique months of the year in which the campaign data was captured is also noted. This month count acknowledges seasonal variations in ocean clarity (Smart 1993), with a higher number of unique months being more representative of an

average profile across the year rather than one heavily influenced by season. We also calculate a “diversity factor” as the product of the cruise count and unique months, which provides a single figure to quantify the diversity of each campaign count.

There are undoubtedly aspects of the data analysis methodology that could be refined and extended, and so the full source code of our analysis has been provided to permit others to reproduce our results and modify our approach (Williamson and Hollins 2023). Our selection of data cleaning criteria could be adapted, as could the boundaries we selected for the classification of campaigns and the setting of depth layers. Directly measured K_d values can be affected by shading of the vessel taking the measurements (Voss et al. 1986), and this is not addressed in the present study. Another factor that was not considered was the influence of seafloor depth. An increase in turbidity could be expected near the seafloor, particularly for coastal waters (Smart 2004), which was not accounted for in our work. Indeed, the fact that coastal areas are likely to be shallower is one potential explanation for the lack of data found for the most turbid coastal water types.

Future studies could look at the variation of chlorophyll concentration and CDOM with depth, as well as exploring the relative combinations of absorption and scattering. WOOD holds data for chlorophyll concentration and CDOM, and that could provide a useful extension to this work by providing some context for the discovered variations in turbidity with depth. Furthermore, additional data within WOOD could be used to understand the constituent components of ocean waters at different depths. The present study has derived a depth classification based on the single parameter K_d , but this value has contributions from absorption and scattering. Different ratios of absorption and scattering can result in the same K_d , which raises the question of whether a given Jerlov water type at the near-surface has the same composition as that same Jerlov water type at depth.

Optical communications and photosynthesis are two areas where our work could be used to simulate efficiencies through the vertical water column in “typical” environments. In such cases, only the near-surface Jerlov water type needs to be known (or set by assumption), and then the clarity at other depth layers can be inferred using our results (Table 2) together with the K_d data for Jerlov water types (Fig. 1, with tabulated data available from Jerlov (1976) and Williamson and Hollins (2023)).

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